CLOUD CONNECTOR

CLOUD CONNECTOR, LLC

Cloud Connector Project

Joint Permit Application Narrative

Submitted to

Louisiana Department of Conservation and Energy

Office of Permit and Compliance

and

U.S. Army Corps of Engineers – New Orleans District, Regulatory Branch

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JOINT PERMIT APPLICATION NARRATIVE

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1.0 INTRODUCTION

Cloud Connector, LLC (Cloud Connector), a subsidiary of Venture Global LNG, Inc., proposes to construct, own, and operate the Cloud Connector Project (Project). The Project will deliver feed gas from the existing pipeline grid in Louisiana to the Venture Global Plaquemines LNG, LLC (Plaquemines LNG) Terminal located in Plaquemines Parish, Louisiana and the Plaquemines Expansion, LLC (Plaquemines Expansion) Terminal which will be located adjacent to the Plaquemines LNG Terminal. The Cloud Connector pipeline consists of 165.1 miles of 48-inch-diameter natural gas pipeline.

The Project also includes associated aboveground facilities, including two compressor stations, one interconnect station, two trap sites, one loop termination site, seven mainline valves (MLVs), permanent and temporary access roads, and barge access routes. The entire Project is located within the U.S. Army Corps of Engineers New Orleans District (COE - New Orleans District), including approximately 100 miles within the designated Louisiana Coastal Zone.

The Project will deliver up to 2 billion cubic feet per day of domestically produced feed gas to the Plaquemines LNG and Plaquemines Expansion liquefied natural gas (LNG) export terminals, for liquefaction and export of LNG to the global market. Plaquemines LNG has requested an uprate from the Federal Energy Regulatory Commission (FERC) and U.S. Department of Energy (DOE) to increase the peak output of the Plaquemines LNG Terminal from 27.2 million metric tonnes per annum (MTPA) to 35.0 MTPA (FERC Docket No. CP26-53-000 and DOE Docket No. 16-28-LNG). Additionally, Plaquemines Expansion plans to build, own, and operate additional liquefaction facilities to export LNG on property adjacent to the Plaquemines LNG Terminal. The Plaquemines Expansion project will have a peak capacity of 31 MTPA (FERC Docket No. CP26-27-000 and DOE Docket No. 25-143-LNG) upon full buildout. Plaquemines LNG's existing pipeline interconnections do not provide the feed gas supply necessary to meet the increased demand for both Plaquemines LNG and the Plaquemines Expansion; therefore, the Cloud Connector Pipeline is proposed to transport the additional feed gas capacity required to supply the Plaquemines LNG and Plaquemines Expansion Terminals with gas for power generation facilities and for increased liquefaction capacity to meet the projected demand.

This Joint Permit Application (JPA) Narrative (Narrative) describes the proposed work, environmental analyses, and impact avoidance and minimization measures associated with the authorizations sought by Cloud Connector individually from the Louisiana Department of Conservation and Energy Office of Permitting and Compliance (LDCE OPC) under the Coastal Use Permit (CUP) regulations in Louisiana Administrative Code (LAC) 43:I.Ch.7 and the authorizations sought by Cloud Connector from the COE - New Orleans District under the Clean Water Act (CWA) Sections 404, of the Rivers and Harbors Act of 1899 Sections 10 and 14.[1]

Construction and operation of the pipeline will primarily result in temporary impacts on waters of the U.S., with permanent impacts associated with development of the associated aboveground facilities (e.g., compressor stations and permanent access roads) and conversion of forested wetlands to non-forested wetland cover types as part of vegetation maintenance of the pipeline right-of-way.

[1] The Rivers and Harbor Act of 1899 Section 14 is commonly referred to as Section 408 as sourced from its codification at 33 USC § 408.

As indicated in Figures 1 and 2, the Project facilities are within the COE - New Orleans District, which can be further divided into those that are located within the Louisiana Coastal Zone and those that are not. The Project enters the Louisiana Coastal Zone at MP 100 and continues to its terminus at the Plaquemines LNG Terminal. The proposed pipeline route and workspace layout within the Louisiana Coastal Zone areas shown on the alignment sheets in Figure Set 3 and Figure Set 4. These sheets also show wetland locations, with data drawn primarily from field delineations, supplemented by desktop delineated wetlands. Desktop wetland delineation methodologies were utilized in locations where access for field delineations could not be obtained, and were developed utilizing a variety of data sources, including nearby field data, recent aerial view photography, topographic maps, U.S. Department of Agriculture hydric soils data, light detecting and ranging (LiDAR) data, and U.S. Fish and Wildlife Service National Wetlands Inventory (NWI) mapping. Topographic route maps are provided in Appendix A; the aerial view maps in Appendix B provide the same route details as the topographic maps but with wetland/waterbody overlays based on field and desktop delineation data. Wetland and Waterbody Crossing Summary Tables are provided in Impact Tables 4 and 5, respectively. The wetland delineation report prepared for the Project is provided in Appendix A.

A pre-application meeting with the COE - New Orleans District was held on April 29, 2026 and with the LDCE OPC on April 30, 2026.

The JPA submittal by Cloud Connector includes:

- LDCE and COE Application Forms
- Narrative – as a complement to the Application Form, the Narrative provides additional details about the Project's principal elements. It describes the work to be conducted by Cloud Connecter, identifies associated wetland and waterbody impacts and provides the necessary needs, alternatives, and justification analyses. It also discusses compensatory mitigation for long-term impacts on wetlands.
    - Appendix A – Wetland Delineation Report
    - Appendix B – Additional Temporary Workspace Locations and Justification
    - Appendix C – Environmental Protection Plan
    - Appendix D – Spill Prevention and Response Procedures Plan
    - Appendix E – HDD Inadvertent Returns Contingency Plan
    - Appendix F – List of Project-Related Permits, Approvals, and Consultations
    - Appendix G – Hydrologic Modification Impact Analysis (HMIA)
    - Appendix H – Permit Drawings (Plats) – depicting proposed work in the Louisiana Coastal Zone

2.0 DEMONSTRATION OF NEED (JPA STEP 8E)

The Project will transport domestically produced natural gas to the Plaquemines LNG and Plaquemines Expansion Terminals for liquefaction, storage, and export to overseas markets. This conversion of natural gas to LNG will promote a liberalized global natural gas trade and greater diversification of global supplies. Customers in markets around the world are importing more LNG from the United States to provide their citizens with a cleaner alternative to coal and other higher-emitting forms of energy. American LNG also provides security and supply diversity to countries seeking to enhance their national security. The purpose of the Project is to create the firm transportation capacity needed to transport the quantity of feed gas required for the Plaquemines LNG and Plaquemines Expansion facilities.

The Project will result in the following benefits to the public interest:

- Stimulation of the local, state, regional, and national economies through job creation;
- Generation of substantial tax revenues and improvements of the United States' balance of trade; and
- Increased economic activity and tax revenues, and increased trade with foreign countries.

These benefits are described in Section 2.3.

In addition to being consistent with the public interest, the Project is consistent with supporting the State of Louisiana's role in advancing national energy policy initiatives in line with Executive Orders 14154 – Unleashing American Energy, Executive Order 14156 – Declaring a National Energy Emergency, and Executive Order 14213 – Establishing the National Energy Dominance Council. These combined executive orders set priorities to utilize the United States' unrealized energy resources domestically and for export sale as a reliable, diversified, and affordable supply of energy. Energy growth, including the increased capacity of natural gas for liquefaction and export, would create jobs and economic prosperity, improve the United States' trade balance, strengthen relations with allies and partners and support international peace and security[2], further demonstrating its need.

2.1 PROJECT PURPOSE AND WATER DEPENDENCY DETERMINATION

The Project's purpose is to connect to the existing transmission pipeline grid in Louisiana to supply domestically-produced natural gas required for operations at the Plaquemines LNG and Plaquemines Expansion facilities adjacent to the Mississippi River, a deep-draft navigable waterway with access to the Gulf of America.

The Plaquemines LNG facility, which is currently under construction and commissioning, will liquefy domestically produced natural gas, store up to 800,000 m^3 of LNG, and export up to 27.2 MTPA of LNG to the global market via deep-draft vessels upon its completion. The

[2] EO 14154 – Unleashing American Energy, *available at*: https://www.federalregister.gov/documents/2025/01/29/2025-01956/unleashing-american-energy; EO 14156- Declaring a National Energy Emergency, *available at*: https://www.federalregister.gov/documents/2025/01/29/2025-02003/declaring-a-national-energy-emergency; EO 14213 – Establishing the National Energy Dominance Council, *available at*: https://www.federalregister.gov/documents/2025/02/20/2025-02928/establishing-the-national-energy-dominance-council.

Plaquemines LNG Terminal has also initiated proceedings with FERC to increase the peak capacity of the facility (see FERC Docket Nos. CP26-53-000 and CP26-27-000).

The Plaquemines Expansion facility, which is a joint project between Plaquemines LNG and Plaquemines Expansion, LLC, will export up to 35 MTPA of LNG to the global market. (See FERC Docket No. CP26-27-000).

Consistent with COE guidance, the Project is to supply domestically produced natural gas to the Plaquemines LNG and Plaquemines Expansion facilities, where it will be liquefied, stored, and subsequently exported as LNG to the overseas global market. Consequently, the Project purpose of the Pipeline is to transport the quantity of feed gas required for the operation of the Plaquemines LNG and Plaquemines Expansion Terminals from natural gas supply points to the Terminal Facilities.[3] Special aquatic sites are prevalent throughout Louisiana and must be crossed by the pipeline in order to provide natural gas to the Plaquemines LNG and Plaquemines Expansion facilities. The Pipeline does require access, proximity to, and siting within special aquatic sites to fulfill its basic purpose.

2.2 LOUISIANA DEPARTMENT OF CONSERVATION AND ENERGY PURPOSE AND NEED

Article IX, Section 1 of the Louisiana Constitution of 1974 sets forth the "public trust doctrine" and imposes "a duty of environmental protection on all state agencies and officials, establishes a standard of environmental protection, and mandates the legislature to enact laws to implement fully this policy."[4] The State and Local Coastal Restoration Management Act (Act) was enacted by the Louisiana Legislature. La. R.S. 49:214.22(1). The Act establishes a coastal management program with LDCE and requires anyone seeking to engage in activities that will significantly impact coastal waters to obtain a Coastal Use Permit from LDCE. La. R.S. 49:214.26; La. R.S. 49:214.30. LAC Title 43, Part I, Ch. 7 fulfills the constitutional requirement by providing guidelines for applicable coastal uses and requirements for permits, including consideration of purpose, need, alternatives, and mitigation. Specifically, the LDCE Coastal Use Guidelines (LAC 43:I.701.C), factors 7 through 13, require applicants to provide information on (among other things) the following:

7. Economic need for and extent of impacts of use on economy of locality;

8. Extent of resulting public and private benefits;

9. Extent of coastal water dependency of the use;

10. Existence of need of use and extent of impacts of use on economy of locality;

11. Extent of impacts on existing and traditional uses of the area and on future uses for which the area is suited;

[3] Similar projects have been found to have a water-dependent purpose as it relates to pipelines supplying natural gas to LNG terminal facilities. *See, e.g.,* FEIS on Cameron Interstate Pipeline Expansion Project (April 2014), *available online at*: https://elibrary.ferc.gov/idmws/file_list.asp; and FEIS on Plaquemines LNG and Gator Express Pipeline Project (May 2019), *available online at*: https://www.ferc.gov/industries/gas/enviro/eis/2019/05-03-19-FEIS/05-03-19-FEIS.pdf.

[4] *Save Ourselves, Inc. v. La. Env. Control Comm'n*, 452 So.2d 1152, 1156 (La.1984).

12. Proximity to and extent of impacts on important natural features such as beaches, barrier islands, tidal passes, wildlife and aquatic habitats, and forest lands;

13. Extent to which regional, state and national interests are served, including the national interest in resources and the siting of facilities in the coastal zones as identified in the coastal resources program.

As public trustee for the coastal management program, LDCE is required to follow the guidelines when determining whether to grant a CUP. As noted above, the Project location is dependent upon the Plaquemines LNG and Plaquemines Expansion facilities' position on the Mississippi River to fulfill the Project purpose. Construction and operation of the Project will result in significant positive economic impacts both in the State of Louisiana and nationally, generating public and private benefits through the generation of tax revenues and job creation, while advancing the United States' energy growth objectives.

2.3 DETERMINATION OF BENEFITS

The Cloud Connector Project will provide a cost-effective source for new domestic natural gas to meet international market demand and customer requirements. The Project will deliver up to 2 billion cubic feet per day (bcf/d) of natural gas to the Plaquemines LNG and Plaquemines Expansion Facilities, where it will be processed and exported as LNG via ocean-going LNG carriers under authorities granted by the U.S. DOE Office of Fossil Energy. Based on recent economic studies of the United States' current and projected supply-and-demand patterns for domestically produced natural gas, the Project will provide benefits that outweigh potential adverse impacts. The main benefits associated with the Project are presented below.

Direct Job Creation/Employment Sustainability

The Project will require approximately 4,500 workers for construction of the Pipeline. Additionally, hundreds of off-site workers will be required to support the Project during construction. Upon completion of construction, the Pipeline will require approximately 30 full-time employees to operate.

National Security and Foreign Relations

By promoting a global, liquid, and robust market for natural gas, the United States will increase economic trade and ties with foreign nations by providing them with access to a reliable supply of LNG. This trade will enhance the national security of the United States by encouraging positive foreign relations and promoting economic prosperity with trading partners.

Economic Benefits

In addition to job creation to support construction and permanent employment during operation, the Project will result in increased tax revenues for the parishes in which the facilities will be located. At the local level, the Project will generate sales tax revenues, as well as ad valorem tax revenues associated with the pipeline.

The Project will also allow natural gas producers within Louisiana additional access to international markets for natural gas via liquefaction and transport as LNG to overseas locations such as Europe and Asia. Louisiana may indirectly benefit financially by receiving royalties as the mineral owner of exported natural gas.

Environmental Benefits

Benefits will be realized from the importation of LNG by foreign countries and its consequent increased availability as fuel for power generation and transportation. LNG is likely to displace existing and future coal, heavy fuel oil, and diesel power generation, and displace heavy fuel oil, diesel, and gasoline for vehicles and marine engines, thereby reducing greenhouse gases, sulfur oxides, nitrogen oxides, and other products of combustion.

3.0 PROJECT FACILITIES DESCRIPTION (JPA STEP 5A)

As discussed in Section 1.0, the Project includes the pipeline and its associated aboveground facilities. Table 3-1 provides the overall land requirements for those Project facilities, as described in Section 3.1.

Table 3-1
Cloud Connector Project
Summary of Project Land Requirements within the Louisiana Coastal Zone

Facilities	Temporary Workspace (acres)	Operational Footprint (acres)	Total (acres)
Pipeline			
Cloud Connector	1,664.12	540.96	2,205.08
Aboveground Facilities			
Compressor Station 1	5.49	67.09	72.58
MLV 1	-	0.34	0.34
MLV 2	-	0.17	0.17
MLV 3	-	0.17	0.17
MLV 4	-	0.17	0.17
Pipe Bridge Foundations	-	0.03	0.03
Trap Site 1	7.4	4.41	11.81
Trap Site 2	1.43	6.51	7.94
Permanent Access Roads	--	7.15	7.15
Temporary Access Roads	61.01	--	61.01
Subtotal	75.33	86.04	161.57
Total	1,739.45	627.00	2,366.65

3.1 PIPELINE

As depicted in the Project's Regional Location map (Figures 1 and 2), the pipeline will consist of 165.1 miles of 48-inch-diameter natural gas pipeline within the COE - New Orleans District, including approximately 100 miles within the Louisiana Coastal Zone.

The construction and operation of the pipeline will require the acquisition of temporary right-of-way and permanent operational easements. The typical upland construction right-of-way will be 150 feet wide with a 50-foot-wide permanent easement maintained after construction. The

typical wetland construction right-of-way will be 125 feet wide with a 50-foot-wide permanent easement maintained after construction. Portions of the pipeline constructed within wetland and marsh environments utilizing push methodologies will be constructed within a 150-foot-wide construction right-of-way with a 50-foot-wide permanent easement maintained after construction. Portions of the pipeline will use barge lay construction methodology within a 300-foot-wide construction right-of-way with a 50-foot-wide permanent easement maintained after construction. Workspace configurations are described in greater detail in Section 5.1.

Additional Temporary Workspace (ATWS) outside of the construction right-of-way will be required for road and waterbody crossings, existing utility line crossings, points of inflection along the route, staging areas, pipeline and high-voltage power line crossings, reverse lay sections, hydrostatic test section breaks, MLVs and other facilities, areas where special construction methods will be implemented (e.g., the horizontal directional drill (HDD) or guided bore drilling methods), and areas where additional space is needed for storage of stripped topsoil. Unless topographic or other factors impose constraints, or where the adjacent upland consists of actively cultivated or rotated cropland or other disturbed land, ATWS will generally be set back 50 feet from the edges of waterbodies and wetlands. When construction is complete, the ATWS will be restored to pre-construction conditions. Appendix B summarizes the locations of ATWS and their purposes.

3.2 ABOVEGROUND FACILITIES

Aboveground facilities, including the compressor station, MLVs, and trap sites are required to operate the pipeline system. Aboveground facilities within the Louisiana Coastal Zone are listed below, including:

- One Compressor Station (MP 70.0);
- Four MLVs (MP 77.3, 50.1, 30.8, 11.8); and
- Two Trap Sites (MP 95.1 and 34.3).

The location and extent of aboveground facilities are provided in Figure Set 5. Grading and Plot plans are provided in Figure Set 11.

3.3 PIPE BRIDGE FOUNDATIONS

An elevated pipeline bridge will be constructed where the pipeline crosses the Oakville to St. Jude Polder Levee, just southwest of the Plaquemines LNG and Plaquemines Expansion Terminals at MP 0.3. A 24 x 24-foot pipe bridge foundation will be installed on either side of the levee. Pipe bridge foundation locations are provided in Figure Set 5, and grading and plot plan details are provided in Figure Set 11.

3.4 ACCESS ROADS

Access roads will be used to transport construction workers, equipment, and materials between the construction work area and public highways/roads. These access roads include some private roads and/or two-track roads that may require minor modification or improvement as well as construction of some new access roads to safely support expected loads. Where needed, the access roads will be widened to a nominal 25 feet. Access road locations are provided in Figure Set 6.

3.5 BARGE ACCESS CHANNELS

Cloud Connector will also utilize barge access channels and approved barge access routes to access the pipeline rights-of-way during construction of the pipeline.

Barge access channels will be established in accordance with the approved drawings and applicable permits. Where dredging is required, the spoil will be side cast adjacent to the channel. The locations of dredge piles will be marked and lighted in accordance with U.S. Coast Guard requirements and applicable permits. Gaps will be left in the spoil piles every 500 feet to accommodate local watercraft and facilitate drainage. Proposed barge access channels are shown in Figure Set 7.

For each barge access route, bathymetry will be assessed to determine the maximum vessel draft allowed. Approved routes will be marked with cane poles to facilitate identification and navigation. Certain routes may be approved for use only at high tide. Signage will be posted as necessary to alert vessel traffic to area restrictions.

4.0 ALTERNATIVES (JPA STEP 11B)

Under the Clean Water Act [CWA, 40 Code of Federal Regulations (CFR) § 230], Congress established a permit program [33 U.S. Code (USC) § 1311; 33 USC § 136] for pollutant discharge, or dredge and fill, into "navigable waters". Under Section 404 of the CWA, a permit from the COE is required prior to discharging dredged or fill material into waters of the U.S. as defined in 33 CFR § 328.3(a) and include a range of wet environments such as rivers, streams (including intermittent streams), and wetlands.

Section 404(b)(1) of the CWA provides that the COE must issue such permits "through the application of guidelines" developed by the United States Environmental Protection Agency (EPA), found at 33 CFR §§ 320.2(f), 320.4(a)(1), 320.4(b)(4), 323.6(a)). These guidelines, referred to as Section 404(b)(1) Guidelines (Guidelines), establish various criteria to be considered by the COE in evaluating alternatives, which dictates that:

> no discharge of dredged or fill material shall be permitted if there is a practicable alternative to the proposed discharge which would have less adverse impact on the aquatic ecosystem, so long as the alternative does not have significant adverse environmental considerations.

The Guidelines also specify that where a permitted activity is proposed that will require access or siting in a special aquatic site, all practicable alternatives to the proposed action that do not involve activities in a special aquatic site are presumed to have less adverse impact on the aquatic ecosystem, unless otherwise clearly demonstrated. The Guidelines define practicable as "a means available and capable of being done after taking into consideration cost, existing technology, and logistics in light of overall project purposes."

To be considered preferable to a proposed action, an alternative, known as the Least Environmentally Damaging Practicable Alternative (LEDPA), must provide a significant environmental advantage, meet the objectives and timeframes of the proposed action, and be practicable. Two alternatives to the proposed action, including the no-action alternative, are analyzed below to assess whether there is a practicable and environmentally preferable alternative to the proposed action.

4.1 NO-ACTION ALTERNATIVE

Under the no-action alternative, the pipeline would not be constructed and associated environmental impacts would therefore be avoided. However, adoption of the no action alternative would preclude the pipeline's receipt of natural gas and the transport of that natural gas to Plaquemines LNG and Plaquemines Expansion for liquefaction at and export from the Plaquemines LNG and Plaquemines Expansion Terminals. Under the no action alternative, failure to construct the proposed facilities would render the increased Plaquemines LNG Terminal liquefaction capacity unattainable and, consequently, it would be nonviable, as would the Plaquemines Expansion Terminal. A nonviable LNG export terminal would be counter to executive orders that set priorities to utilize the United States' unrealized energy resources for export as a reliable, diversified, and affordable supply of energy. As such, the no-action alternative is not a reasonable alternative.

4.2 PIPELINE ROUTE ALTERNATIVES

Cloud Connector identified and reviewed pipeline route alternatives to deliver feed gas from the existing transmission gas network in Louisiana to the Plaquemines LNG and Plaquemines Expansion Terminals. This review led to selection of a preferred route.

The geographic and system selection criteria for major route alternatives were that they intersect at locations that allow interconnects with existing natural gas pipelines near Melville, St. Landry Parish, Louisiana. Cloud Connector reviewed existing natural gas pipelines in the region, focusing on geographic proximity, pipeline age and condition, and the delivery capacity of each system.

The following criteria were considered to meet the Project's objective to connect with the regional natural gas transmission network to supply natural gas to the Plaquemines LNG and Plaquemines Expansion Terminals and for consideration of further refinement:

- Minimization of pipelines' length;
- Maximization of collocation with existing pipeline, electrical transmission line, or other linear right-of-way corridors;
- Minimization of levee, waterbody, wetland, and forest land crossings;
- Minimization of potential impacts on federal lands and other areas of incompatible land use, such as mapped public lands, and wildlife management areas; and
- Avoidance of densely populated areas;

This evaluation of alternatives is based on publicly available information and considers the potential impact of the siting, construction, and operation of natural gas transmission facilities on the environment.

To provide consistency in comparisons, desktop level data were used to assess route alternatives. Information used during the analysis included aerial photographs, United States Geological Survey (USGS) topographic maps, NWI data, Rextag energy mapping data, and public and agency input. The environmental analysis considers quantitative data (e.g., counts, acreage,

or mileage) to compare the alternatives such as total length, amount of collocation, and acres of wetlands.

Cloud Connector identified route alternatives for further refinement and comparative assessment with the proposed route.

4.2.1 Major Route Alternative: MP 34.0 – MP 95.0

The route alternative originates at the proposed Trap Site 2 in Assumption Parish, Louisiana approximately 2.5 miles northwest of Paincourtville. The alternative deviates from the proposed action heading east-northeast for about six miles crossing agricultural land. The alternative then turns southeast crossing about 48 miles of wetland of which 45 percent is collocated with existing rights-of-way. However, the wetland community in the area is dominated by cypress-tupelo.

This route alternative was designed to take advantage of collocation opportunities, crossing fewer levees, and less bottomland hardwood wetlands. However, construction costs for the route alternative represents a six percent increase over the proposed action and would have significant and unavoidable impacts on cypress-tupelo forested wetlands (Table 4-1 and Figure 8a). Therefore, the route alternative is not practicable due to increased cost and does not offer a significant environmental advantage when compared to the proposed route. The route alternative was eliminated from consideration in favor of the proposed route.

Proposed Action (Preferred Route)

The proposed action originates at the proposed Trap Site 2 in Assumption Parish, Louisiana approximately 2.5 miles northwest of Paincourtville. The route heads southeasterly crossing predominantly agricultural lands for 25 miles until reaching Compressor Station 1. As the pipeline exits the compressor station and heads eastward to the south of Thibodaux, the route continues to cross agricultural lands. As the route enters Lafourche Parish, it crosses an area interspersed with mixed bottomland hardwoods, emergent wetlands, and cultivated upland for approximately 20 miles. After crossing the Delaune-Walds Levee System, the route heads northeast crossing agricultural lands for about 4 miles. The route then turns to the southeast heading toward Trap Site 1.

While the proposed route is longer and is collocated for only 33 percent of the route, it has significant environmental advantages over the route alternative. The proposed route was designed to avoid impacts on wetlands and waters of the U.S. to the maximum extent practicable, particularly cypress-tupelo forested wetlands.

Both routes shown in Table 4-1 would be technically and economically feasible. Although the route alternative minimizes the overall area affected, it would significantly increase impacts on forested wetlands, representing a 187 percent difference.

Cloud Connector designed the Project to meet regulatory requirements established to ensure adequate environmental protection. Based on this analysis, the proposed action provides the most effective means of achieving the Project's purpose and need (i.e., providing additional natural gas supply to the Plaquemines LNG and Plaquemines Expansion Terminals for export to overseas markets). Therefore, this alternative was selected as the proposed route.

Table 4-1
Cloud Connector Project
Comparative Analysis for Major Route Alternatives

Feature/Resource	Unit	Proposed Route	Route Alternative
Length	miles	60.8	56.9
Total Land Affected	acres	1,105.3	1,033.5
Collocation with Existing Rights-of-Way			
Collocation	miles	20.0	40.0
Greenfield	miles	40.8	16.9
Overall Corridor Collocation	percent	33	70
Agricultural Lands[a]	miles	23.0	4.8
Wetlands Construction Impacts [b]			
Bottomland Hardwood	acres	237.2	158.0
Cypress/Tupelo	acres	13.0	444.0
Freshwater Emergent Wetlands	acres	193.2	254.5
Freshwater Forested/Shrub Wetlands	acres	24.0	14.3
Estuarine Intertidal Wetlands	acres	0.0	0.0
Riverine	acres	0.8	6.4
Fastlands	acres	239.6	24.0
Levee Crossings[c]	count	10	7
Major Waterbody Crossings[d]	count	1	1
Cultural Resource Sites and Battlefields	count	9	3
State Managed Lands	count	1	0

[a] Source: NLCD Land Cover Database, 2023
[b] Source: FWS, National Wetlands Inventory, 2025
[c] Source: COE, National Levee Database, 2025
[d] Source: Navigable Waters (Research and Innovative Technology Administration's Bureau of Transportation Statistics, 2026)
Note: Acreages are based on a 150-foot-wide construction corridor.

4.2.2 MP 30.8 to Plaquemines LNG Terminal

Cloud Connector reviewed four route alternatives that would locate the southern portion of the pipeline in more open water to minimize impacts on potential high quality coastal marsh areas.

The alternatives start at approximate MP 30.8 and have the same end point at the Plaquemines LNG Terminal. Route Alternative 2 traverses Bayou Perot, Bayou Rigolettes, and

The Pen south of Lafitte, before joining the proposed route at approximate MP 0.6. Route Alternative 3 traverses Bayou Perot and Bayou Rigolettes south of Route 2, crossing the Barataria Bay Waterway and Bayou Dupont before reconnecting with Route Alternative 2 at the Jefferson and Plaquemines Parish boundary. Route Alternative 4 deviates from Route Alternative 3 approximately 3.0 miles east of Bayou Des Amoreux and then crosses Bayou Perot north of Little Lake. In Jefferson Parish, Route Alternative 4 crosses Harvey Cutoff, Bayou Cutler, Wilkson Canal, and Deer Range Canal before joining the proposed route at approximate MP 1.0. These alternatives are depicted on Figure 8b and comparative analysis of each alternative with the proposed route is provided in Table 4-2.

Table 4-2
Cloud Connector Project
Comparative Analysis for Alternatives
MP 30.8 to Plaquemines LNG Terminal

Feature/Resource	Unit	Proposed Route	Route Alternative 2	Route Alternative 3	Route Alternative 4
Length	miles	30.8	28.6	29.7	29.3
Utility Collocation	miles	2.1	0.8	0.8	1.2
FEMA 100-year Floodplains Crossed[a]	miles	30.7	28.6	28.6	29.3
Potential Cultural Resource Features Crossed[b]	count	2	0	0	0
Land Use[c]					
Marsh/Vegetation	miles	4.6	18.8	21.5	18.9
Open Water	miles	26.3	11.2	7.4	10.6
Federal Levee Crossings	count	1	1	1	1
Utility Crossings	count	16	13	8	12
CPRA Project Sites Crossed[d]	miles	21.6	37.4	38.7	29.3
Public Oyster Seed Beds	miles	7.6	0.0	0.0	0.4
Private Oyster Leases	miles	0.0	0.0	0.0	0.0

[a] Source: Federal Emergency Management Agency (FEMA)
[b] Source: Louisiana Cultural Resources Map (SHPO)
[c] Source: NWDI
[d] The Louisiana Coastal Protection and Restoration Authority (CPRA) plans and implements restoration projects to reduce land loss and maintain or restore the coastal communities and habitats in Louisiana.

The proposed route was designed to avoid crossing marshes with vegetation. Compared to Route Alternatives 2, 3 and 4, the proposed route results in no more than a 2.2-mile increase in total pipeline length. The proposed route represents a reduction in the crossing of coastal marsh areas by over 300 percent when compared to the three alternatives. Additionally, the proposed route crosses fewer miles of Louisiana Coastal Protection and Restoration Authority (CPRA) project sites and is co-located for the last 2.1 miles with an existing pipeline system as it enters the Plaquemines LNG Terminal.

While the route alternatives would avoid cultural features crossed as well as oyster seed bed areas located within Little Lake, the proposed route does not significantly affect these resources. Overall, impacts can be adequately mitigated along the proposed route through agency coordination, construction techniques and BMPs, with no significant environmental advantage to warrant adoption of Route Alternatives 2, 3 or 4.

4.3 COMPRESSOR STATION 1 SITE ALTERNATIVES

Cloud Connector proposes to construct a compressor station as part of the Project. Cloud Connector selected the alternative compressor station locations to optimize gas flow hydraulics, to minimize impacts on the environment and landowners, and be located proximate to existing electrical utility services.

Cloud Connector considered the requirements of the needed volume of gas for the location of the proposed compressor station. Cloud Connector completed a hydraulic analysis to determine the optimum horsepower and compression needed to provide up to 2 bcf/d of throughput to the Plaquemines LNG and Plaquemines Expansion Terminals. The results of hydraulic modeling identified that the compressor station must be installed within approximately 2 miles, upstream or downstream, of the optimal location identified (i.e., a 4-mile interval around MP 70.0). The 4-mile interval along the Cloud Connector pipeline allows location-specific information (e.g., land use, environmental resources, engineering) to be considered in the decision of where to install the compressor station while maintaining the level of hydraulic efficiency necessary to meet the flow requirements.

Use of a compressor station site outside of this 4-mile interval would require additional lengths of connecting suction/discharge lateral piping to be constructed, additional compression, and in some cases a route variation may be needed. Additionally, the further the compressor station is situated from the pipeline or the optimal range of possible locations affects both the hydraulic considerations and non-hydraulic considerations such as impacts on additional environmental resources and landowners.

A primary environmental concern with siting a compressor station is noise impact. The U.S. EPA has indicated that a Day-Night Average Sound Level (Ldn) of 55 A-weighted decibel (dBA) protects the public from indoor and outdoor activity interference. Lafourche Parish maintains noise regulations within its Code of Ordinances, Chapter 16, Article IV, Noise. Two limits are defined in the Lafourche Parish noise ordinance for a facility. The first is that a facility's 10-minute L_{10} (sound level exceeded for 1 minute of that measurement) should not exceed 50 dBA at the property line. The second is that a facility's short-duration sound level (sound exceeded for less than 1 minute of the measurement) cannot exceed 65 dBA at any point during the measurement.

Cloud Connector identified potentially suitable land parcels considering specific resource and engineering issues. The criteria used in Cloud Connector's site selection process generally included:

- Availability of at least 100 acres of land to accommodate the construction and operation of the compressor station;
- Availability of and proximity to electrical service;
- Location of residential or other noise-sensitive areas (NSA); and

- Ease of access (e.g., proximity to existing roads).

Cloud Connector evaluated several compressor station sites that potentially would meet the purpose and need of the Project using the criteria described above. Four alternative sites, including the proposed site, were identified as options for the compressor station as depicted on Figure 8c. Table 4-3 provides a quantitative comparison of land use and other environmental features for each of the alternative sites. At one location, a suitable parcel was identified and then removed from consideration because it is outside of the 4-mile interval and the landowner was not interested in selling the property or the property was not otherwise available.

Table 4-3
Cloud Connector Project
Compressor Station Site Alternatives

Feature/Resource	Unit	Compressor Station 1 Alternative 1	Compressor Station 1 Alternative 2[a] (Proposed)	Compressor Station 1 Alternative 3	Compressor Station 1 Alternative 4
Parcel Size[a]	acres	77.5	164.8	97.2	122.9
Land Use	acres				
Agriculture		77.5	0.0	93.5	119.9
Forest		0.0	164.8	0.0	0.0
Developed		0.0	0.0	3.7	3.0
Wetlands[b]	acres				
Bottomland Hardwood		0.0	162.8	19.4	0.0
Riverine		0.5	<0.1	0.0	0.0
Floodplain (1 percent flood hazard)	acres	77.5	164.8	0.0	5.3
Fastlands	acres	77.5	162.9	58.4	0.0
Cultural Resource Sites	number	0	0	1	0
Distance to Nearest NSA	feet	2,360	1,810	1,910	3,509
Distance to Electric Distribution Line	miles	0.3	0.0	2.0	0.7
Distance to Nearest Public Road (paved)	feet	0.0	0.0	3,160	1,580

[a] The compressor station would affect approximately 70.0 acres during operation.
[b] Source: U. S. Fish and Wildlife Service National Wetlands Inventory

Compressor Station 1 Alternative 1

Compressor Station 1 Alternative 1 site was evaluated but after consultation with the landowner, Cloud Connector learned that the parcel was not available for long-term lease or

purchase. Additionally, the Compressor Station Alternative 1 site is about one mile outside of the 4-mile interval which would necessitate additional compression and a larger facility footprint. Therefore, the Compressor Station Alternative 1 site was not considered further.

Compressor Station 1 Alternative 2 (Proposed)

The Compressor Station 1 Alternative 2 site is on a 164.8-acre parcel. Approximately 75.0 acres would be affected during construction of the facility. The parcel is adjacent to an existing electric distribution line, a possible source of electric power, and is adjacent to the south side of Highway 3185 (Figure 8c).

The closest NSA to the proposed Compressor Station Alternative 2 is approximately 1,810 feet north-northeast of the site boundary which has a natural tree buffer to the south, west, and east. The Project will implement mitigation measures (e.g., install blowdown silencers, acoustically rated pipe, and silencers on blower fans) to limit operational noise associated with the Compressor Station Alternative 2 not to exceed an Ldn of 55 dBA at any NSA during full load operation.

While the proposed Compressor Station Alternative 2 would affect more acres of bottomland hardwood wetlands, it has several advantages over Compressor Station Alternative 3 and Compressor Station Alternative 4. The proposed site is available for long-term use or purchase from a willing landowner, is optimally located meeting the hydraulic requirements of the Project, is in fastlands minimizing potential impacts from flood events, has suitable access from a public road, and its proximity to existing utility infrastructure. Additionally, the implementation of noise mitigation measures would minimize impacts on NSAs. Compensatory mitigation will minimize the impacts on wetland functions.

Compressor Station 1 Alternative 3

The Compressor Station 1 Alternative 3 is located at MP 66.5 in an agricultural field and is just beyond 4-mile interval determined by the hydraulic studies. Cloud Connector's hydraulic model shows that additional compression would be needed to achieve optimal operating conditions thereby necessitating a larger compressor station footprint that would also likely increase ambient noise levels. Additionally, the electric distribution line would be twice as long as the proposed site, affecting more landowners and environmental resources. Compressor Station Alternative 3 was determined to not be available for long-term lease or purchase after consultations with the landowner. As such, Compressor Station Alternative 3 does not provide an environmental advantage over the proposed Compressor Station Alternative 2.

Compressor Station 1 Alternative 4

The Compressor Station 1 Alternative 4 is located in an agricultural field 1,580 feet north of Highway 90. It is approximately 6 miles south of the proposed interconnect with the Transco pipeline and is outside the 4-mile interval determined by the hydraulic studies. To achieve optimal hydraulic efficiency a route variation would be needed to achieve the necessary throughput to the Plaquemines LNG and Plaquemines Expansion Terminals. This route would add about 3 miles to the pipeline route, impact approximately 60 additional landowners, and pass through more densely populated residential areas near Highway 90. As such, Compressor Station 1 Alternative 4 does not provide an environmental advantage over the proposed Compressor Station 1 Alternative 2.

5.0 CONSTRUCTION SEQUENCE AND PROCEDURES

The following sections describe the sequence of construction activities and construction procedures related to how the Project will be constructed. Project construction activities will follow the procedures outlined in the Environmental Protection Plan (EPP) provided in Appendix C. This section summarizes the construction methods, potential impacts, and planned mitigation measures to be implemented during pipeline construction to avoid and minimize environmental impacts on sensitive resource crossings such as wetlands and waterbodies.

5.1 CONSTRUCTION RIGHT-OF-WAY LAYOUT

Typical construction layout diagrams are provided in Figure Set 9, Construction Typical Drawings. Figure CC-58860D-PPL-DTL-TRY-00004 depicts the typical upland construction layout for a 48-inch-diameter pipeline system. The construction right-of-way width in non-agricultural uplands will measure 150 feet, with a 50-foot-wide spoil side and a 100-foot-wide working side. The typical construction right-of-way will accommodate temporary storage of trench spoil, the area needed for operation of large equipment, and anticipated construction techniques (including automatic welding operations). These dimensions are necessary to allow for a safe work area due to the large diameter of the pipeline. Cloud Connector will acquire a nominal 50-foot-wide permanent right-of-way.

Figure CC-58860D-PPL-DTL-TRY-00001 depicts the typical upland construction layout in areas where full width topsoil segregation is required (e.g., agricultural areas), an additional 25 feet of temporary construction workspace will be needed to provide sufficient space to store topsoil. The additional 20 feet of temporary construction workspace will be on the working side of the right-of-way in most cases but may be shifted to the spoil side if site conditions on the working side cannot accommodate topsoil storage.

Figure CC-58860D-PPL-DTL-TRY-00028 depicts the typical wetland construction layout for a 48-inch-diameter pipeline system. In wetlands, the width of the construction right-of-way for conventional lay will be 125 feet, with 50 feet on the spoil side and 75 feet on the working side. These dimensions are necessary to allow for a safe work area due to the large diameter of the pipeline and the extra storage space needed due to wet soil conditions.

Figures CC-58860D-PPL-DTL-TRY-00029 and CC-58860D-PPL-DTL-TRY-00030 show the typical push construction layout. In areas where the push method is used, Cloud Connector will use a 150-foot-wide construction right-of-way to accommodate the additional equipment, wider trench, and extensive spoils associated with this construction method, with 75 feet on both the sides of the centerline. Additionally, Figure CC-58860D-PPL-DTL-TRY-00102 depicts the construction workspace layout options for waterbody crossings within areas where the push method is used. Portions of the pipeline installed using the push method will require a 50-foot-wide permanent right-of-way.

Figure CC-58860D-PPL-DTL-TRY-00068 shows the typical barge lay construction layout. In areas where the barge lay method is used, Cloud Connector will use a 300-foot-wide construction right-of-way to accommodate barges and specialized equipment required to install the pipe in shallow open water habitats. Portions of the pipeline installed using the barge lay method will require a 50-foot-wide permanent right-of-way.

Figures CC-58860D-PPL-DTL-TRY-00002, CC-58860D-PPL-DTL-TRY-00003, CC-58860D-PPL-DTL-TRY-00026, CC-58860D-PPL-DTL-TRY-00027, CC-58860D-PPL-DTL-TRY-

00069, CC-58860D-PPL-DTL-TRY-00100, and CC-58860D-PPL-DTL-TRY-00101 depict the various construction layouts associated with collocation of the pipeline with existing pipelines, roads powerlines. Cloud Connector will use a 150-foot-wide construction right-of-way with a 50-foot-wide permanent easement where the route is collocated with existing pipelines and powerlines.

Figure CC-58860D-PPL-DTL-TRY-00043 shows the typical HDD construction layout with a 50-foot-wide permanent easement. Typical HDD pad site dimensions will be 250 feet by 250 feet. Site-specific HDD drawings are provided in Figure Set 10, HDD Plan & Profile Drawings.

5.2 CONVENTIONAL PIPELINE CONSTRUCTION

Pipeline construction using conventional techniques, described below, typically involves the following sequential activities: clearing and grading; stringing, bending, welding, field joint coating, trenching; lowering-in; backfilling; hydrostatic testing and tie-ins; and cleanup, restoration, and revegetation. The pipeline alignment will be identified and surveyed prior to construction. Before construction, the Louisiana one-call system will be contacted so that buried utilities can be identified and flagged by the facility owners to prevent accidental damage during pipeline construction.

Construction techniques will follow the requirements set forth in permit conditions and the EPP, provided in Appendix C. Construction specifications will also adhere to the Spill Prevention and Response Procedures (SPRP) for fuels, hydraulic fluids, and lubricants, and procedures for specialized construction techniques (e.g., HDD), provided in Appendix D. Site-specific HDD Inadvertent Returns Contingency Plans will be prepared and put into place for each individual HDD location prior to the initiation of an HDD. A typical HDD Inadvertent Returns Contingency Plan is provided in Appendix E.

5.2.1 Right-of-Way Surveying

The pipeline alignment will be identified and surveyed before construction. This task will include staking the pipeline centerline, third-party line crossings, workspace limits, and boundaries of wetlands and other environmentally sensitive areas. Sensitive areas to be avoided will be flagged as appropriate, and wetland boundaries will be clearly marked using easily identifiable flagging and/or temporary signage.

5.2.2 Clearing and Grading

Before clearing the construction workspace, appropriate temporary erosion controls will be installed where feasible and in accordance with the EPP. Typically, silt fence or other sediment barriers are positioned along the limits of disturbance where required and across the right-of-way adjacent to wetlands and waterbody crossings. Environmental Inspectors and construction staff will monitor field conditions daily to verify that appropriate erosion and sedimentation control measures are maintained until the construction workspace is fully stabilized. Clearing and grading operations will incorporate procedures to minimize vegetation removal from slopes, wetlands, and banks, and to prevent undue soil profile disturbance.

The construction work area will be cleared to remove trees, rocks, brush, and roots. After clearing, upland sections of the construction right-of-way will be graded as necessary with a bulldozer or similar equipment to create a safe and level work surface. Trees will only be removed when necessary to facilitate safe and practicable construction. Timber and other vegetation

debris may be chipped for use as erosion-control mulch, burned, or otherwise disposed of in accordance with the EPP, applicable federal and local regulations, and landowner requirements. If approved by the landowner, timber, brush, or wood chips may be stockpiled in an accessible location adjacent to the right-of-way for the landowner to retrieve for beneficial reuse.

After clearing, upland sections of the construction right-of-way will be graded as necessary with a bulldozer or similar equipment to create a safe and level work surface, although given the relatively uniform topography of the pipeline route, extensive grading is not anticipated. Generally, machinery will operate on one side of the trench, and excavated materials will be stockpiled on the other.

Fences will be cut and braced along the right-of-way, and temporary gates or gaps will be installed to provide access to the right-of-way. Segregated topsoil will be placed along the right-of-way in a manner that allows for access, material transport, and pipe assembly. In areas where Cloud Connector stores topsoil and spoil on the same side of the right-of-way, sufficient space or a barrier will be left between the piles to allow for returning the subsoil without disturbing the topsoil pile. Generally, machinery will operate on one side of the trench, and excavated materials will be stockpiled on the other side of the trench.

5.2.3 Conventional Lay Trenching

Trenching involves excavating a pipeline ditch and will be accomplished with backhoes and/or similar excavation machinery. Trenching may be performed ahead of stringing activities or after field joint coating, depending on the contractor's execution plan. Trench sides will be sloped, depending on the stability of the existing soils, as required for safety and to minimize topsoil sloughing into the ditch. The construction right-of-way will accommodate necessary ditch and subsoil spoil storage requirements and minimize subsoil/topsoil mixing and sloughing. The trench will be excavated to a sufficient depth to allow a minimum of 3 feet of cover over the pipeline, except for agricultural lands and specified waterbody crossings, where the minimum depth of cover will be 4 feet. The bottom width of the trench will be cut to accommodate the specific diameter of pipe to be installed. The top trench width will vary depending on local soil conditions at the time of construction. Due to the type of soils crossed by the Project, the need for special bedding or blasting is not anticipated.

Crossing existing third-party pipelines will generally require the new pipeline to be buried at a greater depth than the existing pipelines; these areas will be identified and flagged during the pre-construction phase. Trenching near an existing pipeline will proceed only after appropriate field potholing or approved locating methods are completed to confirm the exact location of the pipeline or other foreign utility.

In active cropland, residential areas, or at the landowner's discretion, topsoil is routinely segregated from subsoil during trenching or grading operations and remains segregated during construction to avoid loss through mixing with stockpiled subsoil.

In accordance with the EPP, measures will be implemented to minimize erosion and sedimentation during trenching. These will include measures to minimize the free flow of surface water into the trench and through the trench from upland areas into waterbodies. Erosion control measures will also be implemented as necessary for bank stabilization at waterbody crossing locations.

If trench dewatering is necessary, discharge to the ground is generally permitted where adequate vegetation along the right-of-way functions effectively as a filter medium. In areas adjacent to waterways or areas of minimal vegetation, straw bale filters, filtration bags, or other appropriate measures will be used to limit sediment dispersion. Trench dewatering will be performed in accordance with applicable permit requirements.

5.2.4 Stringing, Welding, and Installation

Stringing involves moving pipe joints into position along the prepared construction right-of-way. Construction of the pipelines will require the use of large side booms to lift and move the pipe, especially for the 48-inch-diameter pipe, including the probable use of Caterpillar 594 and Caterpillar 589 pipe-laying equipment. These units require a considerable width for safe operation. For instance, a Caterpillar 589 with the boom and counterweight retracted requires a minimum width of 15 feet, while the operational width necessary for boom extension and counterweight is about 38 feet. For safety and operational considerations, a 5-foot minimum width will be needed to offset the equipment from the welded pipe and, depending on soil stability, greater distances may be required. The same distance will be required on the ditch side of the welded pipe for worker safety.

In upland areas, trucks and loaders will move the joints from the pipe yards and place them along the construction right-of-way, parallel to the trench line, for subsequent bending if required, line-up, and welding. Stringing activities will be coordinated with the trenching and pipe-laying crews. Stringing trucks will travel along the right-of-way and lay or string the individual pipe sections on temporary supports (skids) along the working side of the trench to prepare for subsequent bending, line-up, welding, joint coating, lowering-in, backfilling, and associated inspection activities. At waterbody crossings, the amount of pipe required for the waterbody crossing will be stockpiled in temporary work areas on one or both banks of the waterbody.

Welding will be performed in accordance with 49 CFR § 192, Subpart E, Welding of Steel in Pipelines, and API Standard 1104. Following integrity inspections, the pipe will be lowered into the trench using sideboom tractors or similar equipment and bedded with padding material (screened native material) if required prior to backfilling. In sections where the push method will be utilized, after the welded pipe string reaches its final location for push work, the floats supporting the pipe string will be removed, and the pipe string will be sunk to the trench bottom.

Cloud Connector plans to use an automatic/mechanized welding process for the majority of the pipeline welds on the Project. Automatic welding operations will be conducted in portable shelters, commonly referred to as “sheds.” The standard width of a shed is 10 feet, and each shed is moved by track-mounted side booms in a leapfrog manner during welding operations. Depending on the size of the vehicle used, movement of the sheds could require a travel width up to about 30 feet. Additional workspace is required for worker safety on either side of the equipment and sheds moving along the construction right-of-way. If there is insufficient room for the movement of the sheds in the normal construction sequence, then welding operations will have to occur in tandem operations, resulting in a slower rate of overall progress.

5.2.5 Coating Application, Inspection, and Repair

To prevent corrosion, the pipeline will be coated in compliance with U.S. Department of Transportation regulations. The pipe joints will be coated (sprayed-on epoxy) at the mill prior to being delivered to the Project area. The ends of each piece will be bare to allow for welding.

Once each weld has been inspected and accepted, the weld area will be field-coated by the coating crew.

5.2.6 Backfilling and Grade Restoration

After the pipe is lowered into the trench and bedded with padding material, the trench will be backfilled with previously excavated material, using barge-mounted track hoes, amphibious equipment, bulldozers, loaders, and/or compactors, depending on the site conditions. Any excess excavated material or components unsuitable for backfill, will be disposed of in accordance with applicable regulations.

In areas where topsoil has been segregated, the backfilling operation will involve returning excavated subsoil to the trench bottom, decompacting stored topsoil by plowing/ripping if necessary, and then placing the stored topsoil over the returned subsoil in the trench. In upland areas, a soil mound crown will be left over the trench to allow for soil settlement unless the landowner requires otherwise. During backfilling, particular care will be taken to minimize erosion, restore the natural ground contours, and restore surface drainage patterns as close to pre-construction conditions as practicable. Upon completion of trench backfilling, topsoil will be redistributed as necessary and the pre-construction soil profile restored across the construction workspace.

5.2.7 Hydrostatic Testing and Tie-ins

The pipeline will be hydrostatically tested before it is put into service to verify its integrity and its ability to withstand the maximum allowable operating pressure. Topography, class locations, construction spread break points, and the availability of test water will determine the length of each test segment. Pipeline test segments will be capped and filled with water. The pipe test section will then be pressurized and hydrostatically tested in accordance with applicable regulations. Any loss of pressure that cannot be attributed to specific factors such as temperature changes will be investigated. Identified leaks will be repaired, and the test section retested.

Hydrostatic test water will be obtained in compliance with Louisiana regulations. Cloud Connector will screen hydrostatic test water intake to minimize adverse impacts on aquatic organisms. Upon completion of hydrostatic testing, the water will be pumped to the next segment for testing or discharged at the test site. Test water will be discharged through energy dissipating devices (e.g., straw bale filter structures) in accordance with the requirements of the Louisiana Department of Environmental Quality (LDEQ) hydrostatic discharge general permit (LAG670000), using methods described in the EPP. Cloud Connector does not anticipate that any chemical agents will be added to the test water.

5.2.8 Clean-up and Restoration

After the completion of backfilling and topsoil replacement across the construction workspace, all disturbed areas will be final graded, and any remaining trash, debris, or unsuitable backfill will be disposed of in accordance with applicable regulations. Subsequently, the workspace will be protected by implementing appropriate erosion control measures, including site-specific contouring and, if necessary, reseeding with an approved seed mix.

5.2.9 Road and Railroad Crossing Procedures

Pipeline construction across major paved highways, railroads, and unpaved roads where traffic cannot be interrupted will be accomplished by using HDD or a conventional bore technique (jack and bore method) as described below. Smaller unpaved roads and driveways will be crossed by open trenching. If an open-cut road requires extensive construction time, provisions will be made for detours or other measures to permit traffic flow during construction.

To complete a bore, two pits will be excavated, one on each side of the feature to be bored. A boring machine will be lowered into one pit and a hole bored to a diameter approximately 2 inches larger than the diameter of the pipe (or casing, if required) at the depth of the pipeline installation. The pipeline section and/or casing will be pushed through the bore to the opposite pit. Once the bore has reached the other pit, the pipeline segment will be tied into the main pipeline sections already installed on either or both sides. If additional pipeline sections are required to span the length of the bore, they will be welded to the first section of the pipeline in the bore pit before being pushed through the bore. With this method, the pipeline will pass under the railroad or roadway with limited disturbance to traffic.

5.2.10 Waterbody and Wetland Crossing Construction Procedures

Crossings of waterbodies and wetlands will be completed in accordance with permit conditions and the EPP. HDD, bore, and push methods will also be used to cross waterbodies and wetlands in addition to the conventional open-cut method. Typical construction layout diagrams for waterbody and wetland crossings are provided in Figure Set 9, Construction Typical Drawings.

5.2.10.1 Waterbody Crossings

Waterbody crossings will be constructed in accordance with federal, state, and local permits. The proposed pipeline crossing will generally require ATWS on each side of most waterbodies to stage construction, fabricate the pipeline, and store materials and trench spoil. These ATWS will typically be located 50 feet from the waterbody's edge to minimize impacts on riparian vegetation associated with the waterbody. In certain situations, such as where agricultural fields or improved pasture lands are present or where site-specific conditions (e.g., a road crossing adjacent to a waterbody) require a shorter set-back, ATWS may be located closer to, but not less than 10 feet from the waterbody's edge.

Cloud Connector will install the pipeline with the amount of cover needed to meet or exceed DOT and permit requirements. Waterbody crossings will generally require at least four feet of cover over the pipeline. In certain areas, such as those where mechanized dredging or natural channel migration may occur, deeper burial may be required to provide sufficient cover over the pipeline.

To facilitate equipment access across the waterbodies and prevent sedimentation caused by construction, Cloud Connector will install temporary equipment bridges across perennial waterbodies that are not avoided via the HDD method. Temporary equipment bridges will be required along the construction right-of-way. Types of equipment bridges used on the Project may include, but are not limited to, culverts, equipment mats, and railroad car bridges. Equipment bridges will be constructed and maintained to allow unrestricted flow and to prevent soil from entering the waterbody. The bridges will be designed to withstand and pass the highest flow anticipated to occur while the bridge is in place. Bridges may include clean rock fill over culverts

or flume pipes, or timber mat structures, that will remain in place until they are no longer needed. Except for equipment used for clearing and bridge installation, which will be allowed one pass through a waterbody, construction equipment on the right-of-way will be required to use the bridges.

Open Cut Method

The open-cut construction method involves trench excavation, pipeline installation, and backfilling in a waterbody without controlling stream flow (wet-ditch open-cut method) or while diverting streamflow (dry-ditch open-cut method). With the wet-ditch open-cut method, the trench is excavated across the stream using track hoes or draglines working within the waterbody, on equipment bridges, and/or from the streambanks. Once the trench excavation across the entire waterbody is complete, a pre-fabricated section of pipe is lowered into the trench. The trench is then backfilled with the previously excavated material. Following pipe installation and backfilling, the streambanks are reestablished to approximate pre-construction contours and stabilized. Erosion and sediment control measures are then installed across the right-of-way to reduce streambank and upland erosion, and sediment transport into the waterbody.

Flume and Dam-and-Pump Construction Methods

Dry-ditch crossing methods involve conventional trenching of channels that are either dry (contain no discernible flow) or not flowing at the time of crossing. A dry-ditch crossing of a flowing waterbody requires the installation of flume or dam-and-pump structures to isolate most of the stream flow from the trench construction. The flume method involves diverting the flow of water across the construction work area through one or more flume pipes placed in the waterbody. After the flume pipes are placed in the waterbody, sandbags or equivalent dam diversion structures are installed in the waterbody upstream and downstream of the trench area. These devices dam the stream and direct the water flow through the flume pipes, thereby isolating the water flow from the construction area between the dams. An excavator reaches under the flume pipe to dig the trench. The flume pipes and dams typically remain in place during pipeline installation until final cleanup of the streambed and banks is completed.

The dam-and-pump method is like the flume crossing method except that pumps and hoses are used instead of flumes to move water across or around the dammed construction work area. The technique involves damming the stream channel, installing a pump upstream of the crossing, and running a discharge hose from the pump across the construction area to a discharge point downstream of the trench line area. Water flow is maintained throughout the dam-and-pump operation until the pipeline is installed and the streambed and banks are restored and stabilized.

Horizontal Directional Drill Method

The HDD method is a process that allows for trenchless construction by drilling a hole beneath a surface feature, such as a waterbody or other unique resource, and installing a prefabricated segment of pipeline through the hole. For each HDD crossing, electric grid guide wires will be laid by hand along the pipeline centerline to create an electromagnetic sensor grid. The grid will be used by the HDD operator to steer the drill head during drilling. No ground disturbance will be required for installation of the guide wires.

To complete each HDD, a drill rig will be placed on the entry side of the crossing, and a small-diameter pilot hole will be drilled along a pre-determined path beneath the surface feature, using a powered drill bit. As drilling progresses, additional segments of drill pipe will be inserted

into the pilot hole to extend the length of the drill. The drill bit will be steered and monitored throughout the process to maintain the designated path of the pilot hole. Once the pilot hole is complete, the electromagnetic sensor grid will be removed and the hole enlarged to accept the pipeline.

Throughout the drilling process, a fluid mixture consisting of water and bentonite clay (a naturally occurring mineral) will be pumped into the drill hole to lubricate the bit, transport cuttings to the surface, and maintain the integrity of the hole. Water for the mixture will be pumped to the drill site through a hose or temporary network of irrigation-type piping or trucked in from another source. The pump intake will be appropriately screened to prevent the entrainment of aquatic species.

The pipeline segment (also called a pullback section) to be installed beneath the surface feature will be fabricated in the ATWS on the exit side of the crossing while the drill hole is reamed to size. The pullback section will be inspected and hydrostatically tested prior to installation. After the hole is completed, the pipeline segment will be attached to the drill string on the exit side of the hole and pulled back through the hole toward the drill rig.

Mechanical vegetation clearing between HDD entry and exit workspaces is not planned, although minimal vegetation clearing by hand between HDD entry and exit workspaces will be required to accommodate the installation of the guide wires. Travel between the HDD entry and exit workspaces will be primarily by pedestrian means. However, in certain instances, light vehicle access (using rubber-tired vehicles only) may be necessary to set up pumps to support the HDD operations, for inspection purposes, and/or to manage any potential drilling mud releases as described below. Ground disturbance due to light vehicle access is anticipated to be minimal. Any vegetation clearing required to accommodate vehicle access will be conducted by hand.

If an HDD crossing is successful, there are little to no impacts on the surface feature being crossed. However, if a natural fracture or weak area in the ground is encountered during drilling, an inadvertent return of drilling fluid to the environment could occur. Depending on the orientation of the natural fracture or substrate, the drilling fluid may move laterally or vertically from the drill hole. Site-specific HDD Inadvertent Returns Contingency Plans will outline the procedures that will be followed at each HDD location to minimize the potential for an inadvertent release of drilling mud and to undertake effective cleanup should a release occur.

Table 5-1 lists the planned HDD crossings of waterbodies along the portion of the Project within the Louisiana Coastal Zone.

Table 5-1
Cloud Connector Project
Planned HDD Crossings Within the Louisiana Coastal Zone

Parish	Begin Milepost	End Milepost	Approx Length (feet)	Feature Crossed
Assumption Parish	99.6	99.9	2,774	Grand Bayou
LaFourche Parish	68.6	68.9	2,374	Local Hurricane Levee - Thibodaux
Terrebonne Parish	68.4	68.6	2,374	Local Hurricane Levee - Thibodaux
	66.9	67.3	2,100	Bayou Terrebonne
	66.0	66.4	2,099	Southern Pacific Railroad & St Louis Bayou
LaFourche Parish	63.4	63.9	2,290	Hollywood Canal / Bayou Cut Off
	52.5	53.1	3,214	North Lafourche LD 4 System
	49.0	49.8	3,911	North Lafourche LD 4 System
	46.1	47.0	4,797	Delaune - Walds
	44.9	45.5	3,181	Forty Arpent Canal
	43.6	44.0	2,000	Bayou LaFourche
	41.6	42.5	5,048	Levee & Forty Arpent Canal
	34.3	35.1	4,144	Levee & Forty Arpent Canal
	33.4	34.2	4,144	Levee & Forty Arpent Canal
	30.8	31.2	1,999	Gulf Intracoastal Waterway
	21.0	21.9	4,740	Rock Wall & Marsh
Jefferson Parish	9.8	10.5	3,919	Bayou Cutter

5.2.10.2 Wetland Crossings

The pipeline will cross forested, scrub-shrub, and emergent wetlands. Trees and shrubs will be cut flush with the surface of the ground and removed from the wetland in preparation for construction. Non-saturated wetlands may also be mowed prior to construction activities. To avoid excessive disruption of wetland soils and to preserve the native seedbank and rootstock within the wetland soils, stump removal, grading, topsoil segregation, and excavation will be limited to the area immediately over the trench line. A limited amount of stump removal and grading other than what is required immediately over the trenchline may also be conducted if required for the safe operation of equipment or the placement of timber construction mats.

During clearing, sediment barriers, such as silt fence and/or staked straw bales, will be installed and maintained adjacent to wetlands and within ATWS as necessary to minimize the potential for sediment runoff. Sediment barriers will be installed across the full width of the construction right-of-way at the base of slopes adjacent to wetland boundaries. Coir logs or straw

bales installed across the working side of the right-of-way will be removed during the day when vehicle traffic is present and replaced each night. Sediment barriers will also be installed within wetlands along the edge of the right-of-way, where necessary, to minimize the potential for sediment to run off the construction right-of-way and into wetland areas outside the construction work area. If trench dewatering is necessary in wetlands, the trench water will be discharged into stable, vegetated, upland areas and/or filtered through a filter bag or dewatering structure. No heavily silt-laden water will be allowed to flow into a wetland.

Construction equipment working in wetlands will be limited to that which is essential to clear the right-of-way, excavate the trench, fabricate and install the pipeline, backfill the trench, and restore the right-of-way. Within wetlands, pipeline construction will typically utilize a necked-down temporary construction right-of-way of 125 feet, to minimize wetland impacts. Equipment being moved through wetlands by other equipment will be limited. A travel lane constructed of timber mats will be installed across most wetlands, where necessary, to prevent soil disturbance and facilitate equipment crossing the wetland. In general, wetlands that are cultivated at the time of construction are treated as agricultural lands and timber mats are not utilized.

In areas of unsaturated soils or standing water, low-ground-weight construction equipment and/or timber construction mats will be used to reduce rutting and the mixing of topsoil and subsoil. In wetlands, the top 6 to 12 inches of topsoil from the trench line will be stripped and stored separately from the subsoil. The trench will be backfilled with this same topsoil following pipeline installation.

Because little or no grading will occur in wetlands, restoration of contours will be accomplished during backfilling of the trench to match the elevation of adjacent wetland elevations. Prior to backfilling, trench breakers will be installed where necessary to prevent the subsurface drainage of water from wetlands. The subsoil will be backfilled first, followed by the topsoil. Timber construction mats will be removed from wetlands once construction activities are complete.

Push or Barge Lay Method

The pipeline will be constructed using a push or barge lay method in wetlands and marshes that are inundated to a point to allow this special method to be utilized. The push method will require a 150-foot-wide construction right-of-way, and the barge lay will require a 300-foot-wide construction right-of-way.

Within the construction right-of-way for the pipeline, the ditch width is anticipated to be a minimum of 32 feet at the top, due to the poorly cohesive soils within the pipeline construction workspace. This width is based on the +/- 10-foot depth of ditch necessary for the installation of the 48-inch-diameter concrete-coated pipe and ditch side slopes of 4:1. Trench spoil bank heights are anticipated to be relatively low because the excavated material lacks adequate unconfined compressive strength. To accommodate the trench spoil placement storage, the need for spoil banks parallel to the push ditch on both sides is anticipated. A 50-foot-wide area will be required on both sides of the push ditch for spoil banks, equipment travel, and reasonable buffer gaps. The requirement for a 150-foot-wide construction right-of-way is due to the combination of the 50-foot-wide push ditch and the two 50-foot-wide areas for spoil banks, equipment travel and reasonable buffer gaps between the edges of the right-of-way, spoil banks, and ditch. This construction right-of-way width will reduce storage pile height and prevent material from re-entering the trench prior to placement of the concrete-coated pipe.

Equipment and materials utilized where the push or barge lay method is implemented will include marsh buggies, airboats, amphibious equipment, tracked equipment, and barges, which will have the least environmental impact given the inundated and saturated conditions that occur where this construction method is used. The contractor will use discretion in choosing the equipment that will create the least ground pressure for the specific application. Construction will comply with the EPP (see Appendix C) and applicable permit requirements.

During construction preparation, construction teams will identify suitable "push sites," or areas that have all-weather access and are preferably on higher ground from which the long segments of pipeline can be assembled and floated/pushed into place. Push sites in open water areas will consist of shallow-draft spud barges connected to provide a working platform. Using barges with anchor spuds eliminates the need for an anchor spread and anchor-handling boats, minimizing the area affected during construction. At these push sites, construction teams will perform several procedures, including pipe assembly, welding, non-destructive testing, joint coating and coating repairs, and installation of floatation apparatus. Once the push sites are established, the right-of-way work will begin.

The pipe joints, which are typically concrete-coated 40-foot lengths, will be transported as needed by barge or truck from the pipe staging area to the push sites. At the push sites, after the pipe joints are welded together, the weld joints are coated, and the floats are attached, the assembled pipe string will be floated out into the cleared right-of-way as part of the pipeline push operation. If necessary, a cable will be attached to the front of the pipe string and pulled from the other end of the right-of-way section to assist the push operation. Once the assembled pipeline segment is in place, the floats will be removed, and the pipe lowered into the trench. Trench backfilling will begin once the pipe is in place. Restoration of wetlands disturbed by pipeline installation where the push or barge method is used will be in accordance with the EPP included as Appendix C.

5.2.11 Levees

Levees will be crossed via HDD or conventional bore, and site-specific construction procedures will be determined based on coordination with local, state, and federal permitting agencies. Federal levees crossings will require consultation with the COE under 33 USC Section 408. Non-federal levees crossings will require consultation with the local levee districts and parish and state flood control agencies. Table 5-2 identifies the levees crossed by the Project. A detailed supplemental application will be provided under separate cover that describes the construction activities located near levees.

Table 5-2
Cloud Connector Project
Levees Crossed Within the Louisiana Coastal Zone

Parish	Milepost	Levee Name	Waterbody Name	Status	Levee District
Lafourche	75.7	Leighton-Morvant-Welcome System	Bayou Boudreaux	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	68.7	Leighton-Morvant-Welcome System	Unnamed Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	52.8	North Lafourche LD 4 System	Unnamed Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	49.6	North Lafourche LD 4 System	Unnamed Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	46.4	Delaune-Walds Levee System	Unnamed Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	45.3	Lockport to Larose Levee System	Unnamed Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	42.3	Forty Arpent-Ludevine Levee System	Forty Arpent Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	34.0	Forty Arpent-Ludevine Levee System	Forty Arpent Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	33.9	Larose NW Private Agricultural Levee System	Petit Bois Bayou	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	33.8	Larose NW Private Agricultural Levee System	Canal Tisamond Foret	Non-federal	North Lafourche Conservation, Levee and Drainage District
Lafourche	32.3	Larose NW Private Agricultural Levee System	Unnamed Canal	Non-federal	North Lafourche Conservation, Levee and Drainage District
Plaquemines	0.34	Oakville to St. Jude Polder	Unnamed Canal	Federal	Plaquemines Levee District

Sources: COE, 2026; ESRI, 2025

6.0 IMPACT SUMMARY AND LAND REQUIREMENTS (JPA STEP 10A AND STEP 11A)

Section 3.0, Project Facilities Descriptions, provides a written summary of pipeline and associated facilities required for the construction and operation of the Project. Table 3-1, presented in Section 3.0, provides the overall land requirements for those Project facilities.

6.1 FASTLANDS

The LDCE defines fastlands as lands surrounded by publicly owned, maintained or otherwise validly existing levees or natural formations that normally prevent activities, not to include the pumping of water for drainage purposes, within the surrounded area from having a direct and significant impact on coastal waters. Areas verified by the LDCE as fastlands or above the five-foot contour may be exempt from Louisiana Coastal Resources Program requirements and would not require a Coastal Use Permit. Approximately 14 miles of the pipeline alignment,

Compressor Station 1, and Trap Site 1 are located within fastlands. A summary of fastlands crossed by the Project is provided in Table 6-1 below.

Table 6-1
Cloud Connector Project
Fastlands Crossed Within the Louisiana Coastal Zone

Parish	Levee District	Milepost In	Milepost Out
Lafourche	North Lafourche Conservation, Levee and Drainage District	72.7	76.0
Lafourche	North Lafourche Conservation, Levee and Drainage District	68.6	72.7
Terrebonne	Terrebonne Levee and Conservation District	65.5	67.0
Lafourche	North Lafourche Conservation, Levee and Drainage District	45.3	46.4
Lafourche	North Lafourche Conservation, Levee and Drainage District	43.8	45.3
Lafourche	North Lafourche Conservation, Levee and Drainage District	42.3	43.7
Lafourche	North Lafourche Conservation, Levee and Drainage District	34.0	34.5
Plaquemines	Plaquemines Levee District	0	0.3

Sources: LDCE, 2025; ESRI, 2025

6.2 WETLAND AND WATERBODY IMPACTS

The Project will result in unavoidable impacts to wetlands and waterbodies. Wetlands and waterbodies were delineated in accordance with the *Corps of Engineers Wetlands Delineation Manual (1987)* and the *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Atlantic & Gulf Coastal Plain Region, Version 2.0 (2010)*. Wetlands and waterbodies were delineated in the field where access was obtained, supplemented by desktop methodologies in areas with access denials and in expansive marsh habitats. Detailed methodologies and results for delineation of wetlands and waters can be found in the Wetland Delineation Report included in Appendix A.

6.2.1 Impact Quantification for the Pipeline

The Project will affect a variety of wetland and waterbody habitats in varying degrees. Project impacts on most wetlands and waterbodies within the pipeline right-of-way will not result in a net loss of wetland acreage; however, impacts on forested wetlands (PFO1 and PFO2) will result in conversion from forested wetlands to emergent wetlands through the removal of trees. Unavoidable impacts on wetlands for the construction of aboveground facilities, including the compressor station, interconnect station, meter stations, trap sites, MLVs, and associated permanent access roads, will result in permanent loss of wetland acreage, due to the placement of fill within the facility footprint. Permanent impacts to perennial and intermittent streams, ponds, and other open water impacts associated with aboveground facilities have been avoided, and impacts on ephemeral streams have been minimized to the maximum extent practicable. Table 6-2 and Table 6-3 provide a summary of wetland and waterbody impacts.

Table 6-2
Cloud Connector Project
Summary of Wetland Impacts Within the Louisiana Coastal Zone

Facility Type	Cowardin Classification	Temporary Impacts (acres)[5]	Wetland Conversion (acres)[6]	Permanent Loss (acres)	HDD Impacts Avoided (acres)
Pipeline					
Cloud Connector	E2EM	67.09	--	--	2.54
	E2SS	2.83	--	--	--
	PEM	437.76	--	--	18.97
	PFO1	--	244.1	--	6.20
	PFO2	--	28.75	--	1.94
	PSS	148.32	--	--	5.87
	Subtotal:	*656.00*	*272.85*	--	*35.52*
Aboveground Facilities					
Compressor Station 1	PFO1	--	5.19	64.02	--
MLV 2	PEM	--	--	0.17	--
MLV 3	PEM	--	--	0.05	--
MLV 4	PEM	--	--	0.17	--
Trap Site 1	PEM	2.14	--	0.38	--
	PFO1	--	1.81	0.21	--
Pipe Bridge Foundation	PSS	--	--	0.01	--
Temporary Access Roads	PEM	4.46	--	--	--
	PFO1	--	0.78	--	--
	PSS	0.25	--	--	--
Permanent Access Roads	PEM	--	--	1.92	--
	PFO1	--	--	1.58	--
	PSS	--	--	0.16	--
	Subtotal:	*6.85*	*7.78*	*68.67*	--

[5] Pipeline impacts to emergent (E2EM and PEM wetlands) and scrub-shrub (E2SS and PSS) wetlands are considered temporary in nature.

[6] Pipeline impacts to forested (PFO1 and PFO2) wetlands are considered wetland conversion impacts, resulting in the conversion of forested wetlands to emergent wetlands.

Table 6-2
Cloud Connector Project
Summary of Wetland Impacts Within the Louisiana Coastal Zone

Facility Type	Cowardin Classification	Temporary Impacts (acres)[5]	Wetland Conversion (acres)[6]	Permanent Loss (acres)	HDD Impacts Avoided (acres)
	TOTAL:	**662.85**	**280.63**	**68.67**	**35.52**

Table 6-3
Cloud Connector Project
Summary of Waterbody Impacts Within the Louisiana Coastal Zone

Facility Type	Flow regime	Temporary Crossing Length Impacts (feet)	Total Area Impacted (acres)
Pipeline			
Mainline			
	Ephemeral	458	1.48
	Intermittent	2,065	5.67
	Perennial	4,668	8.58
	Open Water (E1UB, E1AB, and PUB)	147,983	877.99
	Subtotal:	*155,174*	*893.72*
Aboveground Facilities			
Trap Site 2			
	Ephemeral	--	0.04
Compressor Station 1			
	Intermittent	--	0.91
Access Roads			
	Ephemeral	--	0.01
	Intermittent	--	0.09
	Perennial	--	0.11
	Subtotal:	*0*	*1.16*
	TOTAL:	**155,174**	**894.88**

6.2.2 Hydrological Modification Impacts Analysis

A Hydrological Modification Impacts Analysis (HMIA) was conducted for the Project under LDCE guidance for Level 2 – Intermediate Modifications. Level 2 projects must identify existing surface water discharge patterns, quantity and rate, and affected water quality parameters. The HMIA explains hydrologic modifications resulting from the construction and operation of the

Project and identifies measures to reduce adverse impacts resulting from the modifications. Please refer to Appendix G for the HMIA prepared for the Project.

7.0 AVOIDANCE AND MINIMIZATION OF IMPACTS ON WETLANDS AND WATERBOTTOMS (JPA STEP 11C)

Avoidance and minimization of impacts on aquatic resource areas have been principal factors in Project planning, which has included a thorough review of potential alternatives (Section 4.0) to identify the least environmentally damaging practicable alternative that still meets the purpose and need of the Project.

Construction and operation of the Pipeline will result in temporary and partial wetland impacts. While impacts on waters of the U.S. will be required to construct the Project, Cloud Connector evaluated measures to minimize impacts, to the extent practicable. Cloud Connector evaluated measures to minimize impacts, to the extent practicable. For example, to minimize habitat fragmentation, the pipeline route has been designed to maximize collocation with existing linear right-of-way corridors where practicable. Approximately 45 miles of the pipeline route are collocated within existing linear rights-of-way corridors.

The pipeline construction right-of-way will generally be reduced from 150 feet to 125 feet within wetlands to minimize impacts. Cloud Connector will also utilize the HDD crossing method in seventeen locations to construct the Pipeline to avoid impacts on numerous waterbodies and/or levees (see Table 5-1 for HDDs locations). The HDD method will allow for trenchless construction for installing the pipe beneath the waterbody or other features, which will avoid direct impacts on the waterbody. During HDD crossing construction, an inadvertent return of drilling fluid could occur, discharging drilling fluids to surface waters. Site-specific HDD Inadvertent Returns Contingency Plans will be prepared and implemented for each HDD location before HDD work begins. The HDD Inadvertent Returns Contingency Plan will outline the procedures to follow to minimize the potential for an inadvertent release of drilling mud and effective cleanup should a release occur. If a release were to occur, implementation of the measures in the HDD Inadvertent Returns Contingency Plan will minimize impacts on surface water resources. A typical HDD Inadvertent Returns Contingency Plan is provided in Appendix E.

The Project has been designed to avoid impacts on wetlands or waterbodies using the HDD method where appropriate. Cloud Connector estimates the use of the HDD method will avoid approximately 36.04 acres of wetland impacts within the Louisiana Coastal Zone[7].

Marsh and open water habitats east of MP 30 will be primarily constructed using push and barge lay techniques to minimize impacts on wetlands and waterbodies. The push and barge lay methods reduce impacts because pipe joints are stored and welded at staging areas or on a barge rather than parallel to the trench as in conventional methods, thereby reducing equipment use and traffic flow along the right-of-way. Use of these construction methods, in combination with collocation, minimizes habitat fragmentation and impacts on wetlands and waters.

The proposed pipeline design specifications and routing configuration represent the outcome of a comprehensive alternatives analysis in which multiple system and route alternatives were evaluated to arrive at the least environmentally impactful option that also offers sound construction feasibility and economic viability. Likewise, multiple sites for the pipeline's

[7] Within HDD drill segments, workspace design only shows the permanent easement. Therefore, this is an underestimate of total impact avoidance within wetlands, since open cut construction would also require ATWS.

compressor stations were evaluated using similar selection criteria. In addition, the locations and footprints of temporary workspaces and temporary access roads were optimized to minimize the spatial extent and impacts of these features.

Apart from the long-term conversion of forested wetland to other wetland types, the impacts associated with pipeline installation are considered temporary and, following construction, pre-construction conditions will be restored to the extent practicable. During and following construction, Cloud Connector will verify that temporarily impacted wetlands and waterbodies are restored according to permit conditions and the EPP. Adherence to both the EPP and the SPRP Plan will minimize potential impacts on groundwater and surface water resources.

For wetlands crossed by the pipeline route, the following minimization and site-specific impact mitigation measures will be implemented:

- minimization of vegetation clearing and soil disturbance;
- avoidance of unnecessary vehicular traffic and equipment use;
- installation and maintenance of erosion and sedimentation control devices such as straw bales and silt fences;
- restriction on construction duration to the extent practicable;
- use of timber mats, where appropriate, to create a temporary work surface in wet conditions; and
- use of low-pressure ground equipment or top soiling, where appropriate, in wet conditions to minimize vegetation damage, soil compaction, and rutting.

Areas of construction-related temporary impacts will generally be restored to pre-construction contours and allowed to revegetate. Cloud Connector anticipates that restoration of the temporarily impacted areas will be monitored after one full growing season to satisfy permit requirements. For locations where pre-construction conditions are not achieved according to this schedule, Cloud Connector anticipates working with the regulatory agencies to determine appropriate compensatory mitigation or remediation measures to facilitate restoration.

8.0 COMPENSATORY MITIGATION PLAN (JPA STEP 11D)

Pursuant to both federal and Louisiana state law, mitigation is required to address dredge and fill impacts in wetlands and coastal waters.[8]

Under COE regulations, compensatory mitigation must be used to offset those unavoidable permanent impacts on waters of the U.S. that are authorized through the issuance of COE permits, pursuant to Section 404 of the CWA and/or Sections 9 or 10 of the Rivers and Harbors Act of 1899.[9] The COE regulations set forth three types of compensatory mitigation:

[8] See 33 CFR § 332.3; LAC 43:I. 724.A.

[9] 33 CFR §§ 332.1(a) and 332.2(a)(1).

permittee-responsible mitigation, mitigation banking, and in-lieu fee (ILF) mitigation.[10] When evaluating compensatory mitigation options, the COE will consider what is "environmentally preferable,"[11] and retains the discretion "to determine the appropriateness and practicability of any compensatory mitigation required for DA permits."[12]

Under LDCE regulations, compensatory mitigation is required to offset those unavoidable impacts on state wetlands that are authorized through the issuance of a CUP, pursuant to LAC 43: I.724.A.

The Project has been designed to avoid and minimize adverse impacts on aquatic resources where practicable. Compensatory mitigation for the unavoidable impacts on wetlands (i.e., wetland type conversion and wetland loss) is being prepared in accordance with the Mitigation Rule and CWA Section 404(b)(1) Guidelines to replace the loss of aquatic resource functions. Cloud Connector, through development of a Compensatory Mitigation Plan using the 2008 Final Compensatory Mitigation Plan (U.S. Environmental Protection Agency, 2008), plans to offset the long-term conversion and/or permanent loss of wetlands associated with the permanent loss and/or conversion of wetlands along the pipeline that would require permanent above-grade facilities or be affected by long-term vegetation maintenance of the right-of-way.

To consistently assess wetland impacts and evaluate and determine the most suitable offset type for compensatory mitigation requirements, the COE - New Orleans District has adopted the Louisiana Wetland Rapid Assessment Method (LRAM). The LRAM was developed for the assessment of wetland habitat types within the geographic boundaries of the New Orleans District and uses the natural habitat community types developed by the Louisiana Department of Wildlife and Fisheries' (2009) Louisiana Natural Heritage Program. In total, the LRAM describes 16 natural communities which are condensed into six different classifications. Because the Cowardin (1979) classification system of an individual wetland can represent several natural or altered wetland communities, the LRAM is more suitable for determining the appropriate offset type for compensatory mitigation because it describes and identifies a wetland community based on the natural community types that are more representative of the geographic region represented by the COE - New Orleans District.

In 2008, the EPA and COE issued revised regulations governing compensatory mitigation for authorized impacts to wetlands and other waters of the U.S. under Section 404 of the CWA. These regulations, 33 CFR § 332, also known as the "Mitigation Rule", are designed to improve

[10] "Permittee-responsible mitigation" means an aquatic resource restoration, establishment, enhancement, and/or preservation activity undertaken to provide compensatory mitigation for which the permittee retains full responsibility. 33 CFR § 332.2. "Mitigation bank" means a site, or suite of sites, where resources (e.g., wetlands, streams, riparian areas) are restored, established, enhanced, and/or preserved for the purpose of providing compensatory mitigation for impacts authorized by DA permits. 33 CFR § 332.2. "In-lieu fee program" means a program involving the restoration, establishment, enhancement, and/or preservation of aquatic resources through funds paid to a governmental or non-profit natural resources management entity to satisfy compensatory mitigation requirements for DA permits. 33 CFR § 332.2.

[11] 33 CFR § 332.3(a)(1). According to COE regulations, the district engineer determines the compensatory mitigation to be required in a DA permit, "based on what is practicable and capable of compensating for the aquatic resource functions that will be lost as a result of the permitted activity." In making this determination, the district engineer "must assess the likelihood for ecological success and sustainability, the location of the compensation site relative to the impact site and their significance within the watershed, and the costs of the compensatory mitigation project." *Id*. Moreover, "restoration should generally be the first option considered because the likelihood of success is greater and the impacts on potentially ecologically important uplands are reduced compared to establishment, and the potential gains in terms of aquatic resource functions are greater, compared to enhancement and preservation." 33 CFR § 332.3(a)(2).

[12] 73 Fed. Reg. 19,632 (2008).

the effectiveness of compensatory mitigation to replace lost aquatic resource functions and areas, expand public participation in compensatory mitigation decision-making, and increase the efficiency and predictability of the mitigation project review process. This Mitigation Rule sets forth the following hierarchy concerning the forms and locations of preferred compensatory mitigation:

1. Mitigation bank credits;

2. In-lieu fee program credits;

3. Permittee-responsible mitigation under a watershed approach;

4. On-site and/or in-kind permittee-responsible mitigation; and

5. Off-site and/or out-of-kind permittee-responsible mitigation.

Based on the Mitigation Rule, there is a stated preference to purchase the appropriate LRAM credits from mitigation banks. Given that approved bank credits are established and already evaluated by regulatory staff, and such credits are available ahead of or coincident with project impacts, there is no temporal loss of function. Cloud Connector is evaluating the anticipated permanent conversion and loss impacts associated with the Project and will coordinate with the LDCE OPC and COE with input from state and federal stakeholders to adequately produce compensatory mitigation in accordance with the Mitigation Rule and CWA Section 404(b)(1) Guidelines to replace the loss of aquatic resource functions.

Cloud Connector will provide a detailed Compensatory Mitigation Plan. Cloud Connector is evaluating the availability of wetland mitigation bank credits, and the conceptual Compensatory Mitigation Plan would use mitigation banks to the extent possible, as sufficient wetland credits located along the pipeline route are currently available as identified in the Regional In-Lieu Fee and Bank Information Tracking System (RIBITS).

Wetland and Waterbody Crossing Summary Tables provided in Impact Tables 2 and 3 provide an overall summary of wetland and waterbody impacts. Detailed impact tables are provided in Impact Table 4 for wetlands and Impact Table 5 for waterbodies to fully account for impacts to water resources associated with the pipeline, aboveground facilities, and access roads. The temporary and permanent impacts presented represent material acreage and quantities of temporary and/or permanent fill in wetlands or waterbodies. Wetland impacts are identified by the Cowardin (1979) classification system, with detailed habitat identified for the purposes of mitigation. The work areas and elements identified in the tables include project features required for construction and operation of the Project as described in this narrative and shown within the Permit Drawings included in Appendix H.

9.0 PROJECT-RELATED PERMITS, APPROVALS, AND CONSULTATIONS (JPA STEP 5C)

Appendix F provides a list of required permits, approvals, and clearances relevant to the Project facilities and identifies the applicable federal, state, or local agency or other authority in each case.

9.1 U.S. ARMY CORPS OF ENGINEERS

As part of this Joint Permit Application to the LDCE OPC for activities requiring CUP authorization, Cloud Connector is also seeking authorization from the COE to perform work in or affecting navigable waters of the U.S. subject to Section 10 of the Rivers and Harbors Act of 1899 (33 USC § 403) and to discharge dredged or fill material into waters of the U.S. subject to Section 404 of the CWA (33 USC § 1344). Additionally, an application for federal levee crossings in accordance with Section 14 of the Rivers and Harbors Act of 1899 (i.e., Section 408 authorization) will be provided to the COE under separate cover. The Project will not cross any waterbodies listed on the National Park Service - National Rivers Inventory.

9.2 LOUISIANA DEPARTMENT OF ENVIRONMENTAL QUALITY

Section 401 of the CWA requires that, before any federal agency may issue a license or permit to conduct an activity that may result in a discharge into navigable waters (including the construction or operation of facilities), the applicant must provide the federal agency with a certification from the State that any such discharge will comply with applicable provisions of the CWA. Accordingly, as noted above, Cloud Connector is also seeking a Section 401 Water Quality Certification from the LDEQ.

9.3 LOUISIANA STATE HISTORIC PRESERVATION OFFICE

For the purpose of complying with Section 106 of the National Historic Preservation Act (NHPA 54 U.S. Code § 306108) and its implementing regulations (36 CFR 800); Cloud Connector completed Phase I archaeology survey where property access was granted by landowners. The archaeology survey within the coastal zone resulted in the identification of six newly recorded archaeological sites and two isolated finds. Additionally, seven previously recorded archaeological sites were revisited. The report presenting the archaeological survey results was submitted to the Louisiana State Historic Preservation Office (SHPO) on May 6, 2026. Phase I architectural survey was conducted and identified 43 historic-age resources, one Civil War battlefield, and one property listed on the National Register of Historic Places. The report presenting the architectural survey results was submitted to the SHPO on May 11, 2026. The SHPO comments will be provided upon receipt.

9.4 ENDANGERED SPECIES ACT

9.4.1 U.S. Fish and Wildlife Service

Cloud Connector sent introductory letters to the USFWS, notifying the agency of their intent to develop the Project and initiated review under Section 7 of the Endangered Species Act (ESA). Ongoing communication with USFWS has occurred since that time.

Cloud Connector conducted an initial review of species listed federally or proposed for listing under the ESA with the potential to occur in the Project area, as identified by the USFWS Information for Planning and Consultation (IPaC) system. On April 17, 2026, Cloud Connector submitted documentation of this review to the USFWS Louisiana Ecological Services Field Office to seek technical assistance and request concurrence with the determinations of effect for the eastern black rail, red-cockaded woodpecker, pallid sturgeon, West Indian manatee, monarch butterfly and alligator snapping turtle. The Project route does not cross any designated critical habitat.

Additionally, on February 21, 2025, an Acoustic Survey Plan for the northern long-eared bat and tricolored bat was submitted to the USFWS, which was approved on March 7, 2025. Cloud Connector conducted acoustic surveys in the Project area to determine the presence or absence of bat species listed and proposed for listing under the ESA. The results of this survey and determinations of effect for bat species were provided to the USFWS, within the April 17, 2026 submittal, including acoustic location of tricolored bats at 69 sites and assumed present at a total of 72 sites along the pipeline. Northern long-eared bat was not detected at any of the sites investigated along the pipeline.

Cloud Connector will comply with the requirements of the Bald and Golden Eagle Protection Act. Because of the potential for bald eagles to be nesting in the Project area, Cloud Connector will conduct preconstruction nest surveys to identify active bald eagle nests within 660 feet of the Project area. If active bald eagle nests are found, Cloud Connector will follow appropriate mitigation measures according to the National Bald Eagle Management Guidelines to avoid impacts on individual bald eagles.

9.4.2 National Marine Fisheries Service

Cloud Connector queried the National Oceanic and Atmospheric Administration National Marine Fisheries Service (NMFS) Section 7 mapper and Southeast Critical Habitat map on July 9, 2025, to identify ESA-protected species under their jurisdiction that could occur at the Project. Cloud Conductor assessed potential impacts to federally listed species, and submitted a Section 7 informal consultation request to NMFS on April 14, 2026, with a determination that the *Project may affect, is unlikely to adversely affect* four listed species, including the green sea turtle (South Atlantic [SA] Distinct Population Segment [DPS]; *Chelonia mydas*), Kemp's ridley sea turtle (*Lepidochelys kempii*), loggerhead sea turtle (Northwest Atlantic [NWA] DPS; *Caretta caretta*), and the giant manta ray (*Mobula birostris*).

9.5 MAGNUSON-STEVENS FISHERY CONSERVATION AND MANAGEMENT ACT

Cloud Connector queried the NMFS Gulf Council's Essential Fish Habitat (EFH) Mapper on August 6, 2025, to identify EFH that could occur at the Project. Cloud Connector developed an EFH assessment and submitted an informal consultation request with NMFS for compliance with the Magnuson-Stevens Fishery Conservation and Management Act on April 17, 2026, requesting concurrence on a finding of minimal adverse effects to EFH.

9.6 LOUISIANA DEPARTMENT OF WILDLIFE AND FISHERIES OYSTER RESOURCES

Using available GIS data to evaluate the presence of Oyster Lease Areas indicates that no private Oyster Lease Areas are affected by the Project. The closest private oyster lease is about 2.5 miles south of the pipeline route and will not be affected by construction activities. The pipeline system and barge access routes do not intersect Oyster Lease Areas; however, the pipeline will cross the LDWF Tier II Little Lake Public Oyster Seed Grounds. Public oyster seed grounds are protected by Louisiana Statute § 56:434. To cross a public oyster seed ground, a CUP from the LDCE OPC and subsequent water-bottom oyster assessment surveys will be required. Mitigation measures for public oyster seed areas will be determined in coordination with the OPC and LDWF.

9.7 LOUISIANA DEPARTMENT OF WILDLIFE AND FISHERIES WILDLIFE DIVERSITY PROGRAM

Burns & McDonnell also submitted a request to the LDWF Natural Heritage Program (NHP) on January 10, 2025 to obtain information regarding the known locations of federally protected species, as well as state-listed species protected under the Louisiana Revised Statute (RS) 56:1904 and LAC Title 76, Louisiana Species of Greatest Conservation Need (SGCN), and sensitive habitats within one mile of the Project centerline. Data was received on January 17, 2025. This information was used as a reference for surveys completed to evaluate the Project for the potential occurrence of protected species and their habitats.

10.0 REFERENCES

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